Problem 14 | Problems T4_PhaseDiagrams I.pdf #14 | Invariant reactions

Reading a Hypothetical Diagram with Several Compounds

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 6

14. In the following graph, a hypothetical phase diagram is represented, identify and explain the following: a) The existing phases in each phase field of the diagram. b) The invariant reactions writing the temperature and the reaction's equation, the composition of each phase inside the reaction and its name. c) The congruent and incongruent fusion phases.

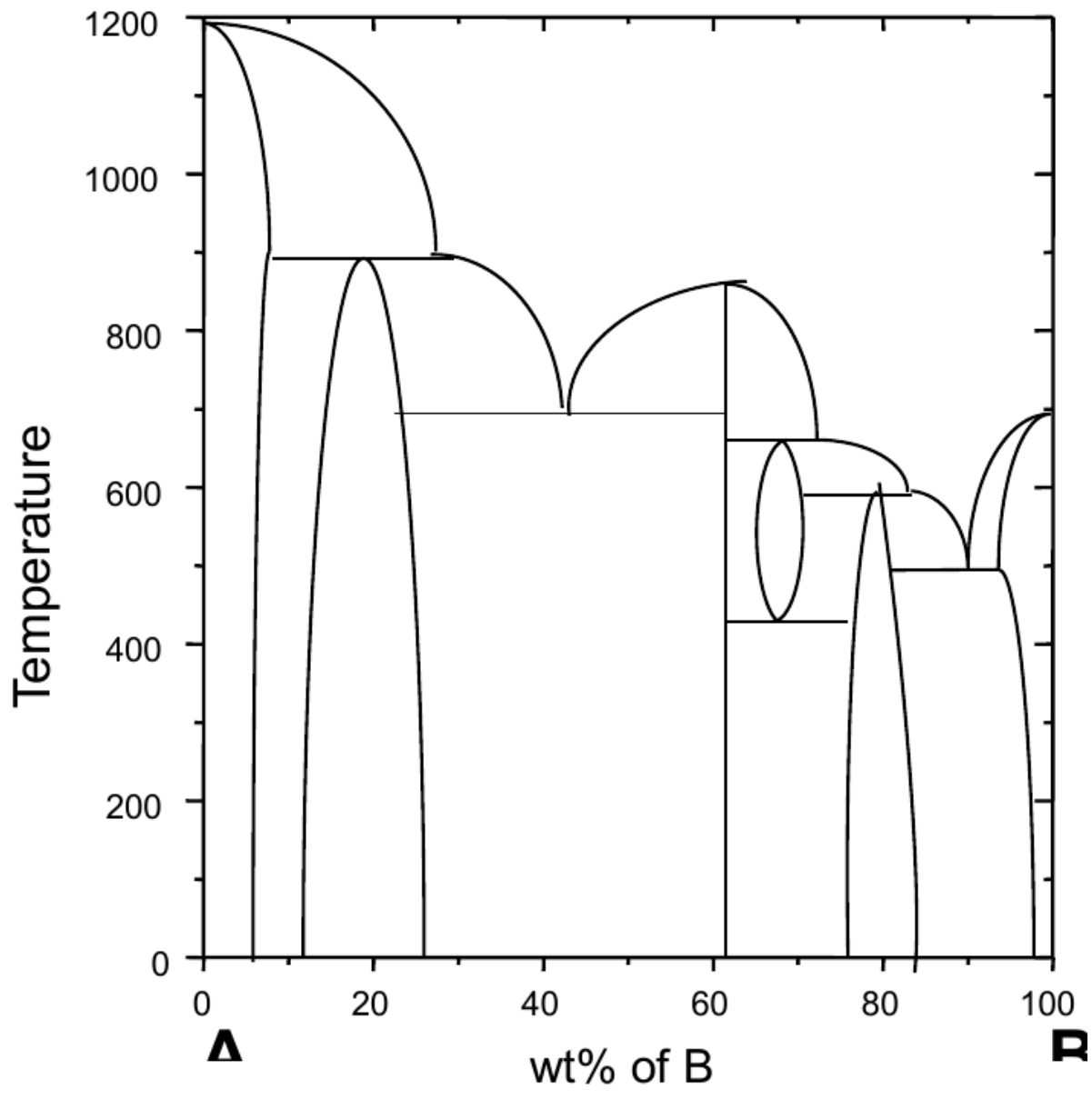


Figure 23. Hypothetical A-B phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 6).

Official Solution (annotated diagram of Solution problems Phase diagrams I.pdf, p. 37; the sheet prints none): β incongruent melting; γ congruent melting; δ incongruent melting; ε incongruent melting

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**; the diagram has no printed values except the axes (wt% B; temperature assumed in °C, since the axis prints no unit), so every number below is a **reading of the figure**, to about ±1 wt% and ±10 °C.
2. Phases are named as in the official solution (p. 37): terminal solid solutions α (A side) and ζ (B side), intermediate phases β, γ (a vertical line: a compound of fixed composition at about 62 % B), δ and ε, and the liquid L.
3. Each horizontal line is an invariant reaction, named by its shape [Session 8 Slide 6]; melting is congruent at a maximum of the liquidus and incongruent at a peritectic line [Session 8 Slide 17].

PHASE 2 · FRAMES & STRATEGY

Frame: composition (wt% B) against temperature. The diagram is read on horizontal lines, from the top down and from left to right; no change of basis is involved.

Method. (a) Single-phase fields are the regions bounded by curved lines that touch the axis or a dome; each two-phase field takes the phases on its two sides along a horizontal tie line [Session 7 Slide 15]. (b) For each horizontal line: the three phases at its ends and middle point, and the shape. (c) Where each intermediate phase ends at the top.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phase fields

Field	Phases
top	L
A side, below the liquidus	α + L, then α
around 20 % B	β; α + β on its left; β + L above the 700 °C line on its right
23-62 % B, below 700 °C	β + γ
43-62 % B, above 700 °C	γ + L
62 % B	γ (vertical line)
62-72 % B, above 660 °C	γ + L
closed field near 68 % B, 430-660 °C	δ, with γ + δ, δ + L and δ + ε around it
62-80 % B, below 430 °C	γ + ε

Field	Phases
around 80 % B	ε, with ε + L above it
81-94 % B, below 495 °C	ε + ζ
B side	ζ + L, then ζ

b) Invariant reactions (on cooling, wt% B)

T (°C)	Reaction	Name
≈ 895	$\alpha\,(8) + L\,(28) \rightarrow \beta\,(19)$	peritectic
≈ 700	$L\,(43) \rightarrow \beta\,(23) + \gamma\,(62)$	eutectic
≈ 660	$\gamma\,(62) + L\,(72) \rightarrow \delta\,(68)$	peritectic
≈ 595	$\delta\,(71) + L\,(84) \rightarrow \varepsilon\,(80)$	peritectic
≈ 495	$L\,(90) \rightarrow \varepsilon\,(81) + \zeta\,(94)$	eutectic
≈ 430	$\delta\,(68) \rightarrow \gamma\,(62) + \varepsilon\,(76)$	eutectoid

c) Melting

- **γ melts congruently** at about 865 °C: it is at the maximum of its liquidus dome, and the liquid formed has its composition.
- **β, δ and ε melt incongruently**: each ends at a peritectic line (895, 660 and 595 °C), where on heating it gives a liquid plus a different solid (β → α + L, δ → γ + L, ε → δ + L).
- The pure components A (≈ 1200 °C) and B (≈ 695 °C) also melt congruently, like any pure metal [Session 8 Slide 17].

PHASE 4 · CHECKS & INTERPRETATION

Checks

- On every eutectic and eutectoid line the parent phase (L or δ) lies **between** the two products; on every peritectic line the product lies between the two reactants, with the liquid at one end [Session 8 Slide 6].
- Phase rule: three phases on each horizontal line, $F = 2 - 3 + 1 = 0$ [Session 7 Slide 13].
- δ is a closed field: it is created by a peritectic at 660 °C and destroyed by a eutectoid at 430 °C, so it exists only between those temperatures.
- **Limiting case**: an alloy with exactly the composition of a line compound (γ) is a single phase, with fractions 1 or 0 and no tie line; every two-phase reading above lies strictly between its tie-line ends.

Interpretation The diagram is a chain of simple diagrams joined at the compounds. A phase that touches the liquid only at a horizontal line cannot exist as a liquid of its own composition, which is the meaning of incongruent melting.

Check against the official solution

Part (c) agrees with the key (β, δ, ε incongruent; γ congruent). The key gives no temperatures or compositions for part (b); they are readings of the figure here. The field labels agree with the annotated diagram of Solution problems Phase diagrams I.pdf, p. 37.